

Edge-Count Separation and Residual Orbit Collisions in a Nonlattice Binary Suspension

Route-A structural certificate C135

Abstract

We refine a symbol-count suspension clock to a directed-edge roof on the full binary shift. An exact four-variable matrix determinant gives the rooted trace law and primitive-orbit product at every period. The nonlattice roof $(1, \sqrt{2}, \sqrt{3}, \sqrt{6})$ separates every distinct directed edge-count sector, including the requested words 000111 and 001011. It remains noninjective on primitive orbits: 001011 and 001101 collide, and a flow-conservation theorem proves that all periodic data are blind to off-diagonal orientation. The construction is source-side and dynamical, not an arithmetic or target-facing factorization.

1 Formal determinant and all-period product

Let Σ be the two-sided full binary shift and assign the ordered edge roof

$$\tau = \begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{3} & \sqrt{6} \end{pmatrix}.$$

The suspension identifies $(x, t + \tau_{x_0 x_1})$ with $(\sigma x, t)$. For formal edge variables put

$$M(x) = \begin{pmatrix} x_{00} & x_{01} \\ x_{10} & x_{11} \end{pmatrix}, \quad \Delta(x) = \det(I - M(x)).$$

Direct calculation gives

$$\Delta(x) = 1 - x_{00} - x_{11} + x_{00}x_{11} - x_{01}x_{10}. \quad (1)$$

For every $n \geq 1$, matrix multiplication yields the exact rooted-word law

$$\mathrm{Tr} M(x)^n = \sum_{\substack{w \text{ rooted closed} \\ |w|=n}} \prod_{i,j} x_{ij}^{N_{ij}(w)}. \quad (2)$$

Expanding $-\log \det(I - M) = \sum_{n \geq 1} \mathrm{Tr}(M^n)/n$ and grouping each rooted word by its unique primitive root and repetition proves, in the total-edge-degree completion,

$$\Delta(x) = \prod_{[\gamma] \text{ primitive}} \left(1 - \prod_{i,j} x_{ij}^{N_{ij}(\gamma)} \right). \quad (3)$$

There are only finitely many primitive necklaces of any fixed total degree, so every coefficient of (3) receives finitely many contributions; the formal product is therefore well defined. Equations (2)–(3) hold at every period.

2 Nonlattice specialization

Substitute $x_{ij} = e^{-s\tau_{ij}}$. Then

$$d_\tau(s) = 1 - e^{-s} - e^{-\sqrt{6}s} + e^{-(1+\sqrt{6})s} - e^{-(\sqrt{2}+\sqrt{3})s}, \quad (4)$$

and (3) specializes to

$$d_\tau(s) = \prod_{[\gamma]} (1 - e^{-s\ell(\gamma)}), \quad \ell = N_{00} + \sqrt{2}N_{01} + \sqrt{3}N_{10} + \sqrt{6}N_{11}.$$

If $h > 0$ is the unique solution of $\rho((e^{-h\tau_{ij}})_{ij}) = 1$, the logarithmic trace and primitive expansions converge absolutely for $\mathrm{Re} s > h$. Indeed, for $\sigma > h$ the positive matrix $M_\tau(\sigma) = (e^{-\sigma\tau_{ij}})$ has spectral radius below one, while absolute values at s are bounded entrywise by $M_\tau(\sigma)$. Hence $\sum_n \mathrm{Tr}(M_\tau(\sigma)^n)/n$ converges and dominates both regroupings. The explicit finite exponential polynomial (4) is entire, so its reciprocal is meromorphic. The fixed cycles $[0]$ and $[1]$ have lengths 1 and $\sqrt{6}$; their irrational ratio rules out a common clock lattice and proves that the roof is nonlattice. It also rules out an imaginary period of the full exponential polynomial: $d_\tau(s + iT) = d_\tau(s)$ for all s would force both $e^{-iT} = 1$ and $e^{-iT\sqrt{6}} = 1$, hence $T = 0$.

3 Exact edge-sector separation

The numbers $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$ form the rational basis of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. For completeness, applying the four independent sign changes $\sqrt{2} \mapsto \pm\sqrt{2}$ and $\sqrt{3} \mapsto \pm\sqrt{3}$ to a rational linear relation and adding with signs isolates each coefficient. Equality of two roof lengths therefore forces equality of their entire directed-edge-count vectors. At period six,

word	$(N_{00}, N_{01}, N_{10}, N_{11})$	rooted multiplicity	length
000111	(2, 1, 1, 2)	6	$2 + \sqrt{2} + \sqrt{3} + 2\sqrt{6}$
001011	(1, 2, 2, 1)	12	$1 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}$

Their difference is $1 - \sqrt{2} - \sqrt{3} + \sqrt{6} \neq 0$. Thus the two words, which have the same symbol population, occupy different edge sectors. Under C130's destination-symbol control $\tau_{ij} = \rho_j$ with $\rho = (1, \sqrt{2})$, both instead have length $3 + 3\sqrt{2}$.

The trace multiplicities also have a direct orbit explanation. A primitive period-six necklace contributes its six rooted rotations. There is one necklace in the (2, 1, 1, 2) sector, giving coefficient 6, and two necklaces in the (1, 2, 2, 1) sector, giving coefficient 12.

4 The residual collision is structural

The primitive words 001011 and 001101 are not cyclic rotations, but both have edge vector (1, 2, 2, 1) and hence the same roof time. They are the first same-edge-count primitive collision, at period six.

This obstruction holds at every period. In any closed binary word, every crossing from 0 to 1 must be balanced by a crossing from 1 to 0; hence

$$N_{01} = N_{10}. \quad (5)$$

Algebraically, write the symbols as integers. Each transition contributes $x_{r+1} - x_r$, which is 1 on 01, -1 on 10, and zero otherwise. Summing around a closed word telescopes to zero and proves (5) without a period cutoff. Every periodic length and determinant coefficient consequently depends on the off-diagonal roof only through $\tau_{01} + \tau_{10}$, never through $\tau_{01} - \tau_{10}$. Ordered formal variables preserve the accounting, but closed-orbit data themselves cannot recover orientation asymmetry. The same quotient is visible directly in (1), where the off-diagonal variables occur only as $x_{01}x_{10}$. Thus replacing (τ_{01}, τ_{10}) by $(\tau_{01} + a, \tau_{10} - a)$, while retaining positive roof values, leaves every periodic length and determinant unchanged.

Progress over C130. C130 separated symbol-population clock sectors but aggregated 000111 and 001011. C135 refines the same exact all-period determinant to complete directed-edge-count sensitivity and separates that pair. It deliberately retains collisions inside one edge sector and proves the exact observable quotient rather than claiming orbit injectivity.

5 Validation and Route-A boundary

The replay prefix contains 2,046 rooted words and 226 primitive cycles through period ten; the theorem has no cutoff. A standard-library checker independent of the producer passes 2,121 assertions, including (5) for every replay word. An independent SymPy reconstruction passes 37 checks; byte replay and all 43 hostile cases (42 repaired-hash plus one stale-hash case) pass.

The strict tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL), overall ROUTE_A_EXPLORATORY. The product in (3) is a symbolic-dynamical primitive-orbit identity, not an arithmetic Euler product. We claim no target divisor or global analytic comparison, prime-like correspondence, arithmetic/local data, Euler factors, root numbers, automorphy, natural self-adjoint or unitary lift, Hilbert–Pólya operator, or Riemann-zero relation. Route B is unauthorized; the literal firewall is NO_BAD_EULER_OR_ROOT_NUMBER.